API Reference

Swap Multi-Input

Copy page

This API returns an executable quote for swapping tokens from multiple origin chains to a single destination chain

POST https://api.relay.link / execute / swap / multi-input

Try it

cURL

cURL

```
curl --request POST \
  --url https://api.relay.link/execute/swap/multi-input \
  --header 'Content-Type: application/json' \
  --data '
{
  "user": "<string>",
  "origins": [
    {
      "chainId": 123,
      "currency": "<string>",
      "amount": "<string>",
      "user": "<string>"
    }
  ],
  "destinationCurrency": "<string>",
  "destinationChainId": 123,
  "tradeType": "EXACT_INPUT",
  "amount": "<string>"
}
'
```

Body

application/json

200 400 401 500

RELAY

user string required

User address that will make the deposit on a given origin chain

origins object[] required

Show child attributes

destinationCurrency string required

destinationChainId number required

tradeType enum<string> required

Available options: EXACT_INPUT , EXACT_OUTPUT

recipient string

refundTo string

amount string

Pattern: ^[0-9]+$

txs object[]

Show child attributes

txsGasLimit number


```
{
  "steps": [
    {
      "id": "deposit",
      "action": "Confirm transaction in your wallet",
      "description": "Depositing funds to the relayer to execute the swap for USDC",
      "kind": "transaction",
      "requestId": "0x92b99e6e1ee1deeb9531b5ad7f87091b3d71254b3176de9e8b5f6c6d0bd3a331",
      "items": [
        {
          "status": "incomplete",
          "data": {
            "from": "0x0CccD55A5Ac261Ea29136831eeaA93bfE07f5Db6",
            "to": "0xf70da97812cb96acdf810712aa562db8dfa3dbef",
            "data": "0x00fad611",
            "value": "1000000000000000000",
            "maxFeePerGas": "12205661344",
            "maxPriorityFeePerGas": "2037863396",
            "chainId": 1
          },
          "check": {
            "endpoint": "/intents/status?requestId=0x92b99e6e1ee1deeb9531b5ad7f87091b3d71254b
            "method": "GET"
          }
        }
      ]
    }
    ],
    "fees": {
      "gas": {
        "currency": {
          "chainId": 8453,
          "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
          "symbol": "USDC",
          "name": "USD Coin",
          "decimals": 6,
          "metadata": {
            "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
            "verified": false,
            "isNative": false
          }
        },
        "amount": "30754920",
```

partial boolean

referrer string

gasLimitForDepositSpecifiedTxs number

originGasOverhead number

slippageTolerance string

Slippage tolerance for the swap, if not specified then the slippage tolerance is automatically calculated to avoid front-running. This value is in basis points (1/100th of a percent), e.g. 50 for 0.5% slippage

Response

200 application/json

Default Response

steps object[]

An array of steps detailing what needs to be done to bridge, steps includes multiple items of the same kind (signature, transaction, etc)

Show child attributes

Example:


```
        "amountFormatted": "30.75492",
        "amountUsd": "30.901612",
        "minimumAmount": "30454920"
      },
      "relayer": {
        "currency": {
          "chainId": 8453,
          "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
          "symbol": "USDC",
          "name": "USD Coin",
          "decimals": 6,
          "metadata": {
            "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
            "verified": false,
            "isNative": false
          }
        },
        "amount": "30754920",
        "amountFormatted": "30.75492",
        "amountUsd": "30.901612",
        "minimumAmount": "30454920"
      },
      "relayerGas": {
        "currency": {
          "chainId": 8453,
          "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
          "symbol": "USDC",
          "name": "USD Coin",
          "decimals": 6,
          "metadata": {
            "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
            "verified": false,
            "isNative": false
          }
        },
        "amount": "30754920",
        "amountFormatted": "30.75492",
        "amountUsd": "30.901612",
        "minimumAmount": "30454920"
      },
      "relayerService": {
        "currency": {
          "chainId": 8453,
          "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
          "symbol": "USDC",
```

RELAY


```
[
  {
    "id": "deposit",
    "action": "Confirm transaction in your wallet",
    "description": "Depositing funds to the relayer to execute the swap for USDC",
    "kind": "transaction",
    "requestId": "0x92b99e6e1ee1deeb9531b5ad7f87091b3d71254b3176de9e8b5f6c6d0bd3a331",
    "items": [
      {
        "status": "incomplete",
        "data": {
          "from": "0x0CccD55A5Ac261Ea29136831eeaA93bfE07f5Db6",
          "to": "0xf70da97812cb96acdf810712aa562db8dfa3dbef",
          "data": "0x00fad611",
          "value": "1000000000000000000",
          "maxFeePerGas": "12205661344",
          "maxPriorityFeePerGas": "2037863396",
          "chainId": 1
        },
        "check": {
          "endpoint": "/intents/status?requestId=0x92b99e6e1ee1deeb9531b5ad7f87091b3d712
          "method": "GET"
        }
      ]
    }
  ]
```


fees object

Show child attributes

breakdown object[]

Show child attributes

balances object

RELAY

Show child attributes


```
      "amountFormatted": "30.75492",
      "amountUsd": "30.901612",
      "minimumAmount": "30454920"
    }
  },
```


details object

A summary of the swap and what the user should expect to happen given an input

Show child attributes


```
  "breakdown": [
    {
      "value": "1000000000000000000",
      "timeEstimate": 10
    }
  ],
  "balances": {
    "userBalance": "<string>",
    "requiredToSolve": "<string>"
  },
  "details": {
    "operation": "<string>",
    "timeEstimate": 123,
    "userBalance": "<string>",
    "sender": "<string>",
    "recipient": "<string>",
    "currencyIn": {
      "currency": {
        "chainId": 8453,
        "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
        "symbol": "USDC",
        "name": "USD Coin",
        "decimals": 6,
        "metadata": {
          "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
          "verified": false,
          "isNative": false
        }
      },
      "amount": "30754920",
      "amountFormatted": "30.75492",
      "amountUsd": "30.901612",
      "minimumAmount": "30454920"
    },
    "currencyOut": {
      "currency": {
        "chainId": 8453,
        "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
        "symbol": "USDC",
        "name": "USD Coin",
```


mintlify

RELAY


```
          "decimals": 6,
          "metadata": {
            "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
            "verified": false,
            "isNative": false
          }
        },
        "amount": "30754920",
        "amountFormatted": "30.75492",
        "amountUsd": "30.901612",
        "minimumAmount": "30454920"
      },
      "totalImpact": {
        "usd": "<string>",
        "percent": "<string>"
      },
      "swapImpact": {
        "usd": "<string>",
        "percent": "<string>"
      },
      "rate": "<string>",
      "slippageTolerance": {
        "origin": {
          "usd": "<string>",
          "value": "<string>",
          "percent": "<string>"
        },
        "destination": {
          "usd": "<string>",
          "value": "<string>",
          "percent": "<string>"
        }
      }
    }
  `
```